SCENE-LEVEL TYPE-CONSISTENT RELABELLING FOR TRANSPARENT GLASS DETECTION

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ABSTRACT

GlassNICOL, a transparent-glass detection dataset, types each glass automatically from a depth-derived height in every view. Type is constant, so adjacent views should agree; the released training labels often disagree, while the hand-labelled test splits almost never do. We repair the labels with the structure that produced them. The scenes are static and only the camera moves, so every view of a glass is a view of one point on the table: we register views through the table plane, carry each box by its base point, chain the views of each glass, and give every box in a chain the chain's most frequent type. Detectors trained on the relabelled set, five (accelerator, seed) cells under a sealed protocol, gain out of distribution in type accuracy and six-class average precision over raw labels and over an adjacent-frame vote; on the standard split six-class average precision does not improve and recall falls about two points. Only type ids change.

Index Terms— transparent object detection, label noise, multi-view consistency, relabelling, glass detection

1. INTRODUCTION

Transparent objects are expensive to label. Depth sensors return little through glass and boundaries are weak, so ground truth usually comes from a proxy the sensor can see. ClearGrasp swaps each object for a spray-painted copy [1]; KeyPose labels keypoints in a few views and propagates them through known camera poses, using an opaque twin only for depth [2]; GlassNICOL photographs each tabletop scene in 25 views, records each scene again with the glasses capped to recover depth, and assigns each box a glass type by matching its depth-derived height to the nearest known glass [3]. KeyPose checks its geometry across views; nothing checks that the type assigned in one view agrees with the next.

We check. After registering adjacent training views and matching boxes, 18.1% of matched box pairs that are the same physical glass carry a different type; the hand-labelled test splits pass the same test at 0.0% and 0.3% (Section 2.1).

The structure that produced the labels can also repair them, and the scene says how. A GlassNICOL scene is static: the glasses stand on the table and only the head camera moves. Every view of a glass is therefore a view of one point of the table plane, its base, and a homography of that plane carries points of it exactly from one view to another. We register views pairwise through the table plane, carry each box by its base point, link boxes whose base points coincide, and merge the links into one chain per glass across the scene; each chain gives its most frequent type to every box in it (Fig. 1, Section 2). Our earlier baseline linked only adjacent frames by box overlap after a partial-affine warp, and a missed frame, a failed registration, or the parallax of a tall glass breaks such a chain, so its boxes get a median of 5.5 votes. Registering through the table plane and carrying base points over two hops gives a median of 10 votes, and on the manual splits the chains merge no two glasses of different type. 12.3% of type ids change and no box moves.

Contributions: (i) a measurement of type-label inconsistency in a released benchmark against its own manual test labels; (ii) a scene-level relabelling method that uses only the dataset's own views, associates through the support plane without using camera poses or depth, and changes type ids alone; (iii) evidence over five (accelerator, seed) cells on two accelerators, under a protocol fixed before training, against both the raw labels and the tracklet vote (Section 3).

Relation to prior work. Repeat-factor sampling [4] and data pruning [5, 6] choose which images the network sees and take every label as given; relabelling changes what a label says. Label noise is usually found from model confidence [7, 8] or the training loss [9], or absorbed by robust training [10], in detectors too [11]; we locate it from the capture geometry, without a model. Majority voting along tubelets or tracklets smooths a detector's output at test time [12, 13, 14, 15], and offboard auto-labelling aggregates detections over a sequence into training labels [16]; we vote on a dataset's released labels, not on detections. Ground-plane homographies associate people across cameras by their feet [17]; we use the table plane the same way, across the views of one moving camera. Label Fusion reprojects one 3D label into every frame [18]; GlassNICOL types each view separately, which is where the disagreement enters. KeyPose discards scans whose keypoints disagree [2]; we keep the frame and correct the label.

2. PROPOSED METHOD

2.1. Type labels disagree on the same object

GlassNICOL assigns type automatically (we call this step the labeller): the height of a depth cluster above the table plane, measured in the capped pass, is matched to the nearest of six known glass heights (5, 9, 11, 13, 17.5 and 22 cm) within 3 cm; the heights and that tolerance are constants of the released labelling code rather than of the paper [3]. Height is measured per view, but type is constant for the object, so one glass should carry one type across the 25 views of its scene. We take the 2,280 consecutive frame pairs of the training scenes, register frame f to $f{+}1$ by background features, carry each box through the registration and match it to the box it overlaps in $f{+}1$ at intersection over union (IoU) above 0.5; pairs whose registration fails are matched unwarped. Of the 2,775 matched pairs, 502 (18.1%) carry a different type (Fig. 2), 87% of them between glasses of neighbouring height.

The dataset authors likewise use manual test annotations to reduce noise from automatic labelling [3]. Those splits yield 0 flips in 93 matched pairs on the standard test, at most 1 in 316 (0.3%) on the out-of-distribution (OOD) test. The same labeller also places two

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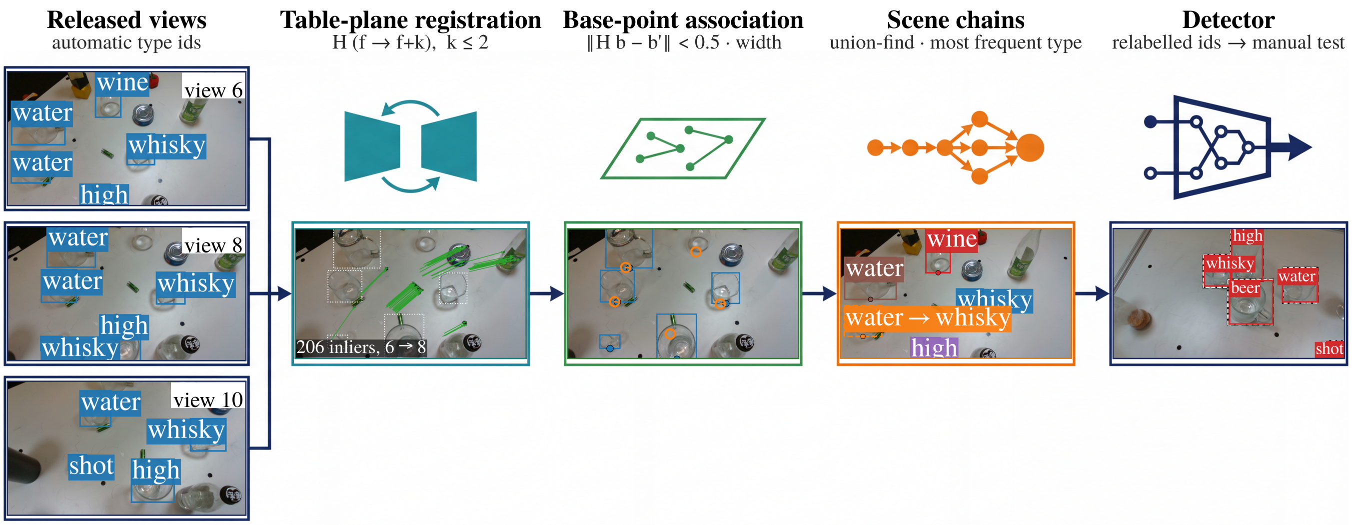

Fig. 1. Overview. Views of a static scene are registered through the table plane by background features; each box is carried by its base point, which lies on that plane, and linked to the box it lands on; union-find merges the links into one chain per glass, whose most frequent type replaces every type id in it; the relabelled set trains a detector scored on the hand-labelled splits. Photographs: views 6, 8 and 10 of one training scene with released boxes and types; inliers of the view 6 to 8 homography as displacement vectors; base points of view 6 carried into view 8 (open) and linked to those of view 8 (filled); view 6 with one colour per chain and a dashed box where the vote replaced the type; a scene-vote detector on a manual OOD test frame, manual boxes dashed, in a frame where every manual box is detected with the correct type and nothing else. Photographs and overlays are real; the stage symbols are schematic.

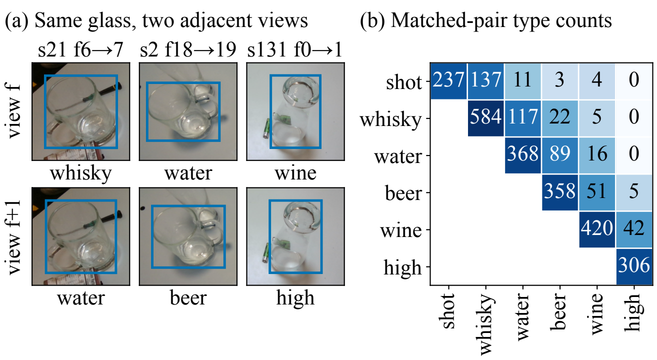

Fig. 2. Automatic type labels are not constant on one object. (a) The same glass in adjacent training views f and $f+1$, cropped from released frames with the released box and type. (b) Released types of the 2,775 box pairs matched across adjacent views after background registration, each unordered pair counted once; 502 (18.1%) disagree, 87% of them between neighbouring heights. Manual test labels flip 0/93 and 1/316.

boxes on one glass (IoU above 0.5) in 51 of 2,375 training frames (44 of 53 such pairs with different types); the manual splits contain none.

The misreads mostly join neighbouring heights and mostly run downward. Treating the views of one glass on an adjacent-frame chain (the tracklets of Section 2.3) as repeated readings and estimating the labeller's confusion by expectation–maximisation [19], a water glass is read as a shorter type in 18.5% of views and as a taller one in 4.7%, a beer glass 15.9% against 0.9%, a wine glass 11.0% against 0.8%, a highball 11.9%; only the shot glass, which has no shorter type, is misread upward, as whisky in 8.9% (Table 1).

Table 1. The labeller's confusion, estimated from adjacent-frame view chains by expectation–maximisation: percentage of views in which a glass of the true type (column) is read as each type (row). Misreads of every type but the shortest mostly go toward shorter glasses.

Read as	shot	whisky	water	beer	wine	high
shot	**91.1**	9.4	4.0	0.4	0.7	–
whisky	8.9	**87.5**	14.5	3.0	1.2	–
water	–	2.3	**76.8**	12.5	1.3	–
beer	–	0.8	2.6	**83.2**	7.7	1.2
wine	–	–	2.1	0.7	**88.3**	10.7
high	–	–	–	0.2	0.8	**88.1**

2.2. Association through the table plane

Frames f and $f+k$ of a scene, $k \leq 2$, are registered by background features: up to 1,500 ORB (oriented FAST and rotated BRIEF) features [20] are extracted outside the label boxes, which are dilated 1.2× and masked out, matched under a 0.75 ratio test, and a homography is fitted by RANSAC (random sample consensus) [21] at a 3-pixel threshold and kept when at least 30 matches and half of them are inliers. The background is the table, so the homography is that plane's and carries points lying on it exactly. We take a glass's base point to be the bottom-centre of its box, an approximation the tolerance below absorbs; the rest of the box moves with parallax. The release annotates a base point per glass for the training split only, not for the hand-labelled splits where we check the association, so we read it off the box. Each box of f is carried into $f+k$ by its base point and linked, one-to-one in order of distance, to the box whose base point lies within 0.5 of the mean box width. Union-find merges links across the scene into chains, one glass seen in several views, under one constraint: a chain holds at most one box per frame, and a link that would violate it is refused (64 of the links proposed by the

Table 2. Association ablation on the training split: the first row is the earlier partial-affine, adjacent-frame baseline; the other rows use the homography registration and vary the linking cue (warped-box overlap or base point) and the hop count. Chains; median votes per box; boxes with three or more votes (%); type ids changed (%); manual-split checks on the standard (Std.) and OOD splits (first row: flips among matched adjacent-view pairs, Section 2.1; other rows: disagreeing boxes).

Association	Chains	Votes	$\geq$3	Chg.	Std.	OOD
adjacent overlap (tracklet)	1,561	5.5	71	8.4	0/93	1/316
overlap, 1 hop	1,115	8	82	9.2	0/167	0/351
overlap, 2 hops	1,012	9	85	10.3	0/167	0/351
base point, 1 hop	893	9	88	10.9	0/167	2/351
base point, 2 hops (ours)	786	10	91	12.3	0/167	0/351

4,097 registered pairs).

2.3. Vote, coverage and validity

Each chain takes its most frequent type (a plurality); ties are left unchanged. Only the type id changes: no box moves, no image is added or removed, and roster and hyperparameters are identical to the raw-label arm of Section 3. The association runs on all 95 training scenes (the detector roster below uses 70 of them; the other 25 stay out of training): 4,097 of 4,465 frame pairs register; 4,336 boxes fall into 786 chains with a median of 10 views per box, 3,953 boxes (91%) are voted on by three views or more, 535 boxes (12.3%) change type and 140 in tied chains keep their label. Table 2 separates the ingredients: the adjacent-frame overlap vote of our earlier baseline, which registered 1,748 of 2,280 pairs by a partial-affine warp, gives 1,561 chains, 878 of them a single box, a median of 5.5 votes per box, and changes 364 boxes (8.4%). Under the homography registration, overlap linking over one hop already lengthens the chains; the base point and the second hop each add votes, and only their combination gives the most votes with no merged glasses of different type on either manual split. That baseline is the tracklet arm of Section 3. The validation split is left as released: checkpoint selection uses generic average precision (AP), so type ids never enter it.

Validity. Manual types are constant per object, so on the hand-labelled splits every within-chain type disagreement is a merged pair of glasses. The association above produces 0 of 167 disagreeing boxes on the standard split and 0 of 351 on the OOD split, with median chain lengths of 10 and 15 (163 of 167 and 346 of 351 boxes in chains of three or more). The hop count and tolerance were chosen by this check on a grid of two to four hops and four tolerances before any scene-vote detector was trained, taking the loosest setting that disagrees nowhere on either split, so the zero above is an in-sample value: at the chosen tolerance, three and four hops leave four disagreements on the standard split and two hops none; a one-hop run made afterwards leaves two on the OOD split. The manual splits never enter training.

Scope. A vote over views removes disagreement that varies by view and inherits whatever is constant across views, such as the class-share difference of Section 4; the released receipts score each pair.

3. EXPERIMENTAL RESULTS

3.1. Setup

Data. The official GlassNICOL head-camera split [3]: 2,375 training images from 95 scenes, 125 validation images, and hand-labelled standard and OOD tests (66 and 75 images, 3 scenes each). The training roster, fixed before this experiment, uses 70 of the 95 scenes (1,750 images).

Detector and regime. YOLO11n [22] (2.59M parameters [23]), 640×640 inputs, batch 64, AdamW [24] at $\mathrm{lr}_0 = 10^{-3}$, cosine decay over 300 epochs. All arms train under the same repeated-exposure regime: the 1,538 training images whose transparent and chalk-coated captures register by the same background matching appear twice, 3,288 slots and 15,600 optimiser updates, and the arms differ only in training type ids: raw, tracklet vote, scene vote. Seeds 1337, 3407 and 4567 run on accelerator A (an H200 GPU) and seeds 1337 and 5678 on accelerator B (an RTX PRO 6000 GPU); a cell is one (accelerator, seed) set of the three arms, an arm being one label variant on the shared roster. On A the raw and scene-vote runs of a cell share a GPU and the tracklet run used another; on B all share one.

Scoring. The checkpoint with the best validation generic AP is scored at confidence floor 0.001 with non-maximum suppression (NMS) at IoU 0.7: COCO (Common Objects in Context) box AP over IoU 0.50–0.95 [25]; generic AP collapses the six types to one. Within each image, GT boxes in ascending annotation-id order greedily take the unused prediction with maximum IoU (≥ 0.5, confidence ≥ 0.25), irrespective of type. Pairwise type accuracy is the correctly typed fraction on boxes localised by both compared arms; recall is the localised fraction per arm.

Existing methods compared. Two standard answers to label noise are trained as arms under the same protocol. Label smoothing [26] keeps the released ids and smooths the classification target of the assigned anchors by 0.1. Detector-confidence relabelling replaces a type id when a detector disagrees with confidence at least 0.5, judged by a model trained on the scene-disjoint half of the roster that excludes that box; it is the confident-learning idea [7, 27] at one global threshold, flipping rather than pruning, and changes 22.5% of the ids on that roster against our 11.7%.

Protocol. The comparison was preregistered before any scene-vote detector was trained: the three arms, the five cells, the checkpoint rule, the four metrics above, and four predictions, a positive mean type-accuracy gain over the raw labels on both splits (P1), an OOD gain exceeding the tracklet vote's (P2), recall and generic AP within two points (P3), and a positive OOD gain in at least 4 of 5 cells (P4). The sealed protocol, every evaluation receipt and the code are at `https://github.com/HeunSeungLim/glassnicol-scene-relabel`. Every contrast is read as a mean over cells with the count of positive cells beside it [28].

3.2. What the labels change

Table 3 pools the seeds per accelerator; the per-cell table with its intervals is released with the code. Two contrasts hold across cells. Out of distribution, six-class AP rises by +7.1 points over the raw labels in 5 of 5 cells (across-cell 95% t-interval [+1.8, +12.3]) and by +6.4 over the tracklet vote ([+0.6, +12.2]), which the tracklet vote itself did not move (+0.7). On the standard split localisation recall falls in every cell, by 2.0 points on average ([-3.7, -0.3]). The type-accuracy gains are larger but less certain: out of distribution the scene vote gains +9.2 points over the raw labels, positive in 4 of 5 cells, with per-cell paired bootstrap intervals over 2,000 draws that resample test images, above zero in 3 cells, reaching it in 1 and below it in 1; resampling whole scenes instead widens them until 8 of 10 include zero and an across-cell interval of [-5.7, +24.1] (cell standard deviation 12.0); over the tracklet vote it gains +4.0 (4 of 5). On the standard split type accuracy changes by +0.8 over the raw

Table 3. Raw, tracklet-vote and scene-vote training labels on the official test splits (manual labels), mean ± standard deviation over seeds per accelerator A or B (number of seeds in parentheses; with two seeds the deviation has one degree of freedom). Six: six-class AP; Gen: generic AP; Rec: localisation recall; Type: type accuracy on boxes localised by all three arms, a different box set from the pairwise contrasts; all in percent.

		Standard				OOD			
Accel. (seeds)	Labels	Six	Gen	Rec	Type	Six	Gen	Rec	Type
A (3)	raw	68.9±3.9	84.0±2.2	89.6±1.2	86.1±4.2	53.8±7.1	66.1±0.9	77.9±4.1	79.9±7.3
	tracklet vote	64.6±4.6	84.8±3.0	88.6±1.2	86.3±3.0	54.0±4.6	67.3±1.0	81.0±0.8	82.6±2.3
	scene vote	61.9±2.7	81.6±3.5	87.6±1.2	82.6±2.0	60.2±3.2	65.9±1.4	77.1±4.1	83.8±9.4
B (2)	raw	60.1±1.6	84.4±2.2	92.5±1.3	77.5±2.8	53.5±0.4	67.4±1.5	82.3±2.0	68.6±5.6
	tracklet vote	67.0±2.8	87.2±0.7	89.5±3.8	83.9±0.7	54.8±7.4	66.9±2.9	78.8±1.4	76.4±7.9
	scene vote	65.9±2.6	83.8±2.5	90.4±3.4	84.3±5.2	61.5±1.5	66.8±2.3	80.3±1.6	84.9±0.1

Table 4. Comparison with existing answers to noisy labels, six-class AP on the official test splits: mean over the 5 (accelerator, seed) cells, the change from the released labels before rounding, and the number of cells above them. Every row trains the same detector on the same roster, budget and seeds, three cells on accelerator A and two on B; the rows differ only in the training type ids, or, for label smoothing, in the classification target. The two external rows were added after the sealed protocol and are post hoc.

	Standard		OOD	
Training labels	AP	vs raw	AP	vs raw
Released labels	65.4	—	53.7	—
Label smoothing	68.7	+3.3 (3/5)	53.1	−0.5 (3/5)
Detector confidence	62.2	−3.1 (1/5)	38.2	−15.5 (0/5)
Adjacent-frame vote (earlier)	65.6	+0.2 (3/5)	54.3	+0.7 (2/5)
Scene vote (ours)	63.5	−1.9 (2/5)	60.7	+7.1 (5/5)

labels (3 of 5 cells positive) and −1.8 over the tracklet vote (1), while six-class and generic AP change by −1.9 and −1.7 (2 and 1 of 5 positive). The one cell that loses on both splits is seed 4567 on A, by eight and ten points; every other gains out of distribution by six or more.

Preregistered predictions. P1 held: the mean type-accuracy gain over the raw labels is positive on both splits, +0.8 and +9.2. P2 held: out of distribution the scene vote's gain exceeds the tracklet vote's gain of +5.0; pairwise type contrasts use different common-box sets and need not add. P4 held: 4 of 5 OOD cells are positive. P3 did not hold: on the standard split localisation recall falls by 2.0 points, just past the two-point bound, and generic AP by 1.7; out of distribution both stay within it. Scoring the last epoch instead (post hoc, rows released with the code) gives type accuracy −0.6 on the standard split, so P1 holds only under the preregistered rule, and +11.3 out of distribution; recall moves +0.5 and −1.7, so the OOD gain holds under both rules while the standard-split recall loss does not. Under that rule the tracklet vote also gains out of distribution, six-class AP +7.0 in 5 of 5 cells, but the scene vote still leads it by +4.1 points in 5 of 5 ([−0.2, +8.3], against [+0.6, +12.2] under the preregistered rule): the lead survives the rule change, but its interval now includes zero.

3.3. Comparison with existing methods

Table 4 sets the two standard answers to label noise, and our earlier vote, beside ours. Label smoothing leaves the detector where the released labels leave it, −0.5 points out of distribution and positive in 3 of 5: absorbing the noise does not recover a wrong type id. Detector confidence moves the labels the most and the scores furthest the wrong way, −15.5 points out of distribution in 0 of 5 cells. It rewrites 22.5% of the ids, and 81.8% of those disagree with the type the geometry votes for: a detector trained on the released labels is not an independent witness. Under the preregistered rule the scene vote alone gains out of distribution in every cell, +7.1 points in 5 of 5, against +0.7 in 2 for the adjacent-frame vote. At threshold 0.8 (post hoc, accelerator A only) it changes 12.8% of the ids, about as many as our vote, and still falls 11.2 points out of distribution in all 3 cells while beating us on the standard split: fewer edits soften the loss without turning it into a gain. Smoothing is the better choice on the standard split, ahead of us in every cell, so the case for relabelling is the out-of-distribution one. The last epoch keeps the order: +0.7 for smoothing, −4.2 for detector confidence, +7.0 for the tracklet vote and +11.1 for ours in 5 of 5.

4. DISCUSSION AND LIMITATIONS

The vote does not bring class shares closer to the tests: the shot:whisky ratio is 0.58 in raw labels and 0.51 after the vote, against 0.79 and 1.83 in the tests. Decoding the chains under the confusion of Table 1 moves the ratio no closer; separating the labeller's contribution from the staging's needs the per-box height, which the release omits.

The association is only as good as the table: frames without enough background features stay unlinked, and the 51 frames with two boxes on one glass are voted twice.

One seed on A loses on both splits and the type-accuracy mean sits inside the seed spread, so six-class AP carries the claim; the tracklet contrast rerun on B for seed 1337 moved by 10.9/14.3 type-accuracy points, a caution for single-cell results. Checkpoint selection by validation generic AP ($r^2 = 0.04$ with test six-class AP) is shared across arms; only labels change.

5. CONCLUSION

GlassNICOL's automatic labels disagree on the same glass in 18.1% of adjacent-view pairs; its hand labels almost never do. Voting over a scene's views changes 12.3% of type ids without moving a box, and under the preregistered rule ours alone gains in every cell out of distribution, +7.1 six-class AP.

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